PROBLEM STATEMENT

The rapid growth of e-commerce platforms has led to a significant increase in transactional data, including product sales, pricing, quantities, and customer interactions. As the volume of this data continues to expand, traditional data processing systems become inefficient in handling such large-scale datasets, resulting in slow processing, limited scalability, and delayed analysis.

E-commerce datasets consist of multiple attributes such as product details, categories, prices, quantities, and transaction dates. With increasing data size and complexity, extracting key business insights—such as identifying top-selling products, calculating total revenue, and analyzing sales trends—becomes a challenging task.

The lack of efficient data processing mechanisms makes it difficult for organizations to perform timely analysis and make informed decisions.

To address these challenges, this project proposes the use of the Hadoop ecosystem, including HDFS and MapReduce, to efficiently process and analyze large-scale e-commerce data in a distributed manner.

Therefore, there is a need for an efficient and scalable data processing approach that can handle large volumes of e-commerce data and support effective analysis.